Experiment-2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

This instruction screen introduces the Flash Number Memory Game clearly and simply. Chunking is applied by presenting information in short, structured sections so users understand the task without cognitive overload.

Clear and Sequential Flow

The instructions explain the task step-by-step, guiding users on what to do before the game begins. Users are informed that a large number will be shown briefly and must be memorized carefully.

Logical Grouping of Information

Task Overview → Explains the goal of memorizing a number.
Memorization Rule → Emphasizes focusing on digit groups rather than individual digits.
Recall Instruction → Informs users that they must select the correct number later.

Visual Hierarchy and Design Elements

A bold title highlights the game purpose.
Minimal text and spacing improve readability.
A clearly visible “START” button indicates progression to the game phase.

Simplicity and Clarity

The instructions are concise and easy to understand, preparing users for the memory challenge.
INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The **Memory Recall Task** instruction page effectively utilizes chunking in the following ways:

FRAME 2:

MEMORIZATION PHASE:

Analysis of the Flash Number Memorization Screen

This screen represents the memorization phase of the Flash Number Memory Game. Users are shown a large multi-digit number for a limited time.

Purpose of the Screen

This phase focuses on visual memory encoding, where users observe and mentally group digits using chunking strategies.

Key Elements and UI Components

Large Displayed Number

A 10-digit number is shown prominently at the center of the screen.
The large font ensures readability and focus.

Countdown Timer

A visible timer indicates how long the number will remain on screen.
This creates urgency and encourages quick memorization.

Chunking Strategy

Users naturally group digits into smaller chunks (e.g., pairs or triples), making the number easier to remember.

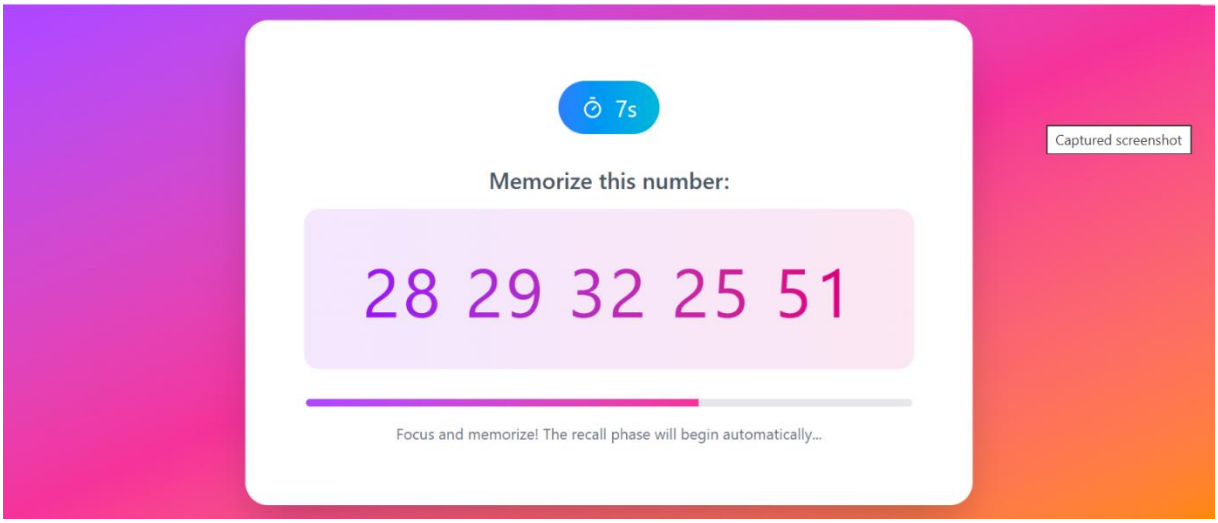
How This Phase Works

Users focus on memorizing the number before the timer ends.
Once the timer finishes, the game automatically transitions to the recall phase.

Cognitive Benefits

Chunking reduces memory load and improves short-term retention.

CHUNKING PHASE:



FRAME 3:

RECALL PHASE:

Analysis of the Flash Number Recall Screen

This screen tests the user's memory by requiring them to recall the previously shown number.

Purpose of the Screen

This is the memory retrieval stage, evaluating how effectively users remembered the number.

Key Elements and UI Components

Number Selection Interface
Multiple number options are displayed as choices.
Some options match the original number, while others act as distractors.

User Interaction

Users select the number they believe matches the memorized sequence.
A "SUBMIT" button confirms their choice.

How This Phase Works

Users rely on chunked memory rather than recalling each digit individually.
Submitting the answer leads to the result screen.

Cognitive and UX Benefits

Tests recall accuracy.
Evaluates the effectiveness of chunking on number memory.
Simple interaction keeps the focus on memory performance.

RECALL PHASE:

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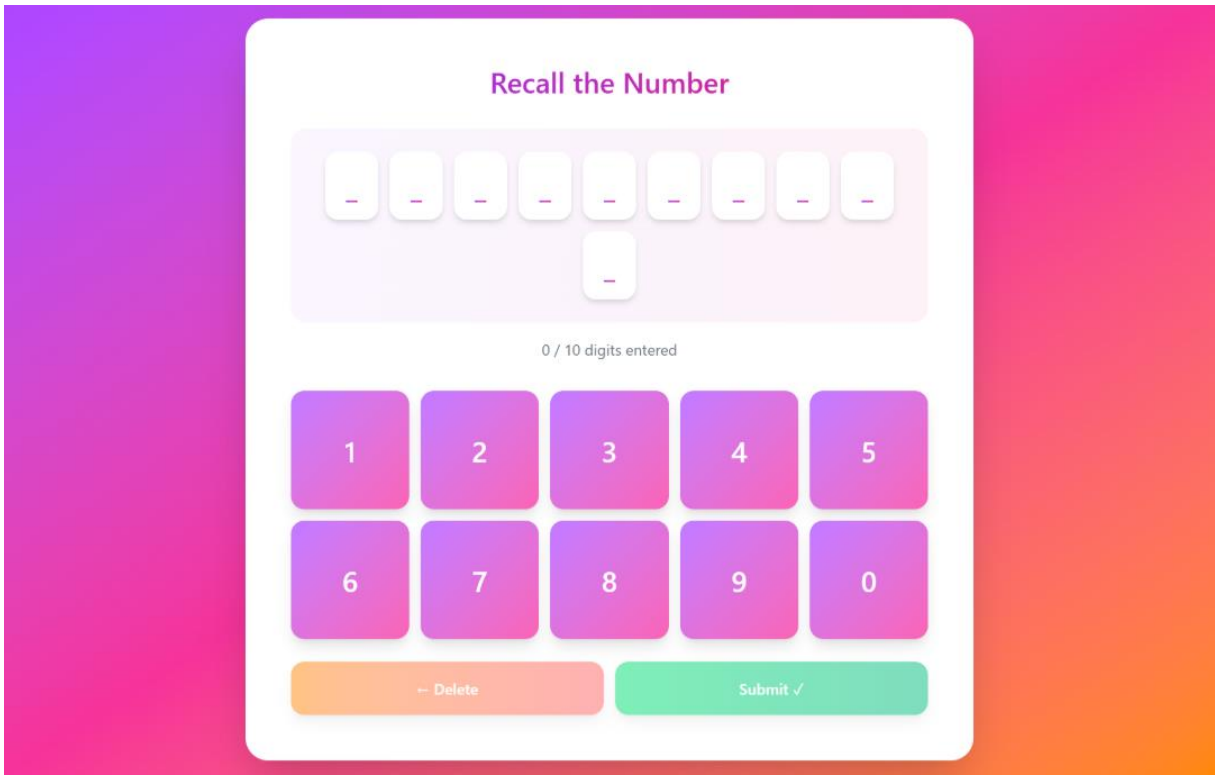


FRAME 4:

RESULT PAGE:

Analysis of the Flash Number Game Result Screen

This screen displays the outcome of the user's recall attempt.

Purpose of the Screen

Provides immediate feedback on memory accuracy.
Allows users to continue or retry the game.

Key Elements and UI Components

Score Display
Shows whether the selected number is correct or incorrect.
May include accuracy feedback.

Action Buttons

Retry → Allows the user to attempt the game again.
Continue → Proceeds to another round or level.

How This Phase Works

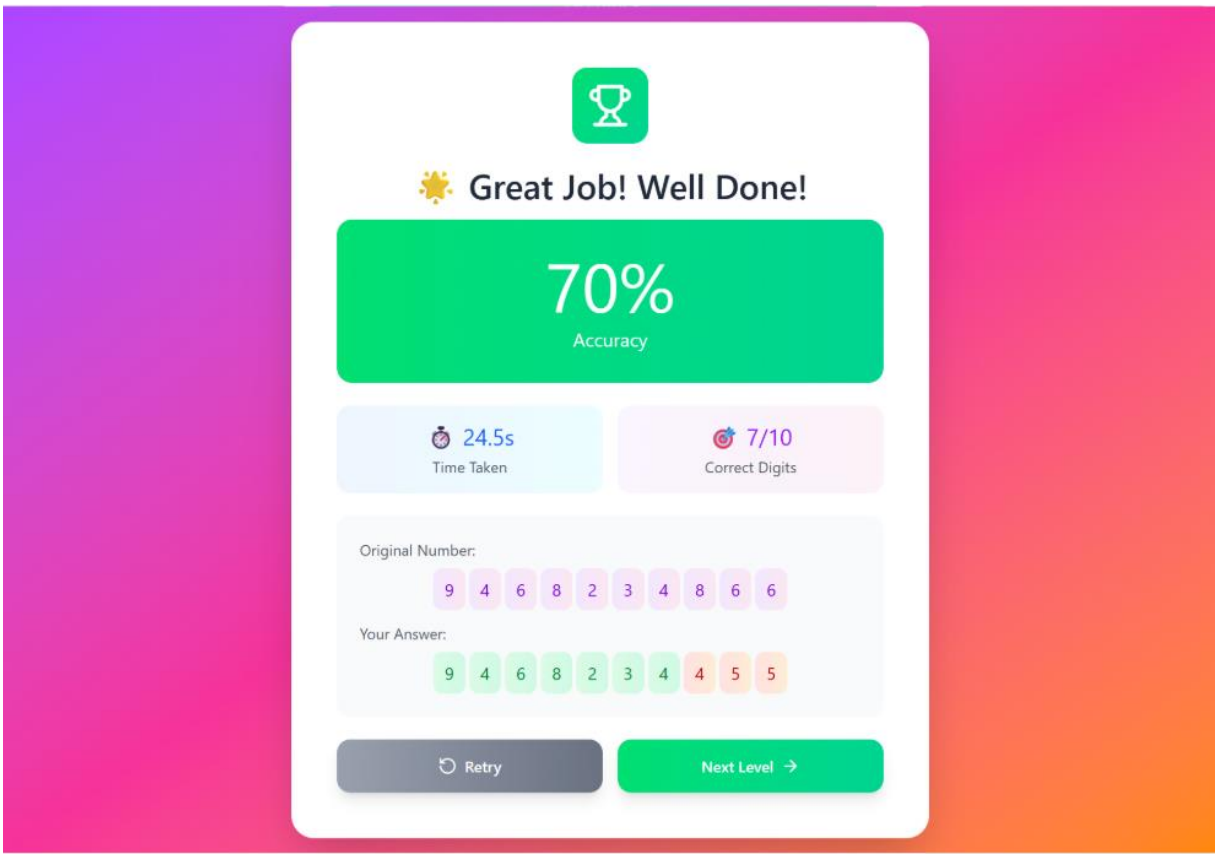
The game compares the user's selection with the original number.
Feedback is displayed instantly.
Users choose their next action.

Cognitive and UX Benefits

Immediate feedback reinforces learning.

Encourages repeated play to improve memory skills.

RESULT PAGE:



PROTOTYPE LINK:

https://www.figma.com/make/vZPI7qFfjbaiS5NSyuU3ZB/Interactive-Number-Flash-Game--Copy-?t=ifpmWXUSE55FAkOo-1